

Run-Pass Options: Plays That Are Two Plays

How one snap carries a run and a pass, why the quarterback picks after the play has started, and why NFL RPOs look nothing like college ones

First-and-10 on the opponent's 40, early in the second quarter. The offense is in the shotgun with three receivers bunched to the right and one alone on the left. At the snap the whole offensive line lurches to the right, run blocking, and the quarterback sticks the ball into the running back's stomach. Everything about it says *inside run*, and the linebacker on the left side believes it: he takes two quick steps forward to fill his gap. Then the quarterback pulls the ball back out, turns, and throws to the receiver on the left, who has run eight yards upfield on a slight angle inside and is standing exactly where that linebacker was a second ago. Twelve yards, first down. The broadcast analyst says "RPO" the way people say "algorithm," as if the word explained itself. So: was that a run or a pass? **It was both, on purpose.** The offense called a run and a pass on the same snap and let one defender's two steps decide which one it would run. This chapter is about plays that are two plays: what they are, why they work, why the NFL's version looks different from college's, and how to see the decision coming.

You'll be able to...

- Say what makes a play an RPO, and tell it apart from play-action (the run is fake) and from the option (both choices are runs).
- Find the defender an RPO is built to put in conflict, usually the apex or overhang defender or a linebacker, and predict give or throw from what he does.
- Tell a pre-snap RPO (decided by counting the box) from a post-snap RPO (decided by reading a defender), and draw the bubble, glance, slant and stick RPOs and the pop pass.
- Explain the NFL's one-yard ineligible-downfield rule, how it differs from college's three yards, and why it forces NFL RPO throws to be quick and short.
- Name the defense's main answers, and say what each one gives back.

i You'll need

- **Option Football:** the **option** principle (leave one defender unblocked on purpose and read him), the **read key**, the mesh and its ride, **give / keep**, the zone read, and the **scrape exchange**.
- **Play-Action, Bootlegs, and Screens:** **play-action** is a pass play dressed as a run: the back fakes, the line mostly pass-blocks, and the throw was always the plan.
- **A First Look at Defense:** the **box** and the **box count**, light and loaded boxes, hat math, the **apex** alignment and the **overhang defender**, man and zone coverage, one-high and two-high shells.
- **Zone Running:** **inside zone**, combo blocks, and linemen climbing to the linebackers at the second level.
- **The Quick Game:** the slant, stick, bubble and now screens, and receiver leverage. **Passing Fundamentals:** the conflict defender.
- **The Twenty-Two:** who is an eligible receiver (only the players at the ends of the line and the backs) and who is not (the five interior linemen).

How to read the diagrams. Offense is blue circles, defense orange squares; *left* and *right* mean the offense's left and right. Q is the quarterback and R the running back; X is the receiver alone on the left, Y the tight end, H the slot receiver and Z the wide receiver on the right. LT, LG, C, RG and RT are the offensive linemen. Defensive linemen are labelled by technique number (03-01): a 5 sits on the tackle's outside shoulder, a 3 on the guard's, a 1 shades the center, a 7 sits on the tight end's inside shoulder. M and W are the Mike and Will linebackers, \$ is the nickel back, CB a cornerback, SS and FS the safeties. Short bars end blocks, thin arrows are routes and paths, thick dark arrows are ball carriers. On the read diagrams, the **dashed green line from the quarterback is his eyes**, a **dashed dark-blue arrow** is the run branch, a **dashed brown arrow** is the pass branch, and a **dashed red arrow** is a defender's possible choice. An **orange ring** marks the defender being read; in the animations a **green ring** follows the ball.

One snap, two plays

A **run-pass option**, or **RPO**, is a play on which the offensive line blocks a run while one or more receivers run pass routes, and the quarterback decides which of the two to use. He decides either before the snap, by counting defenders, or just after it, by watching

one defender while he holds the ball in the running back's belly. If the defense has set itself to stop the run, he throws. If it has set itself to stop the pass, he hands off.

That sounds like every play: of course you run against a pass defense and throw against a run defense. The difference is *when* and *by whom* the choice is made. On an ordinary play the coordinator chooses, from the sideline, some forty seconds before the snap, on a guess about what the defense will do. On an RPO the coordinator calls *both* plays and hands the choice to the one person on the field who can see what the defense actually did, at the last possible moment. Coaches call a call like this a **packaged play**: one call that bundles a run with one or more passes, plus rules for picking. Every RPO is a packaged play; the older, broader term also covers packages with a run, a quarterback keep and a pass, which you will meet near the end of the chapter.

An RPO sits between two plays you already know, and the easiest way to understand it is to put the three side by side.

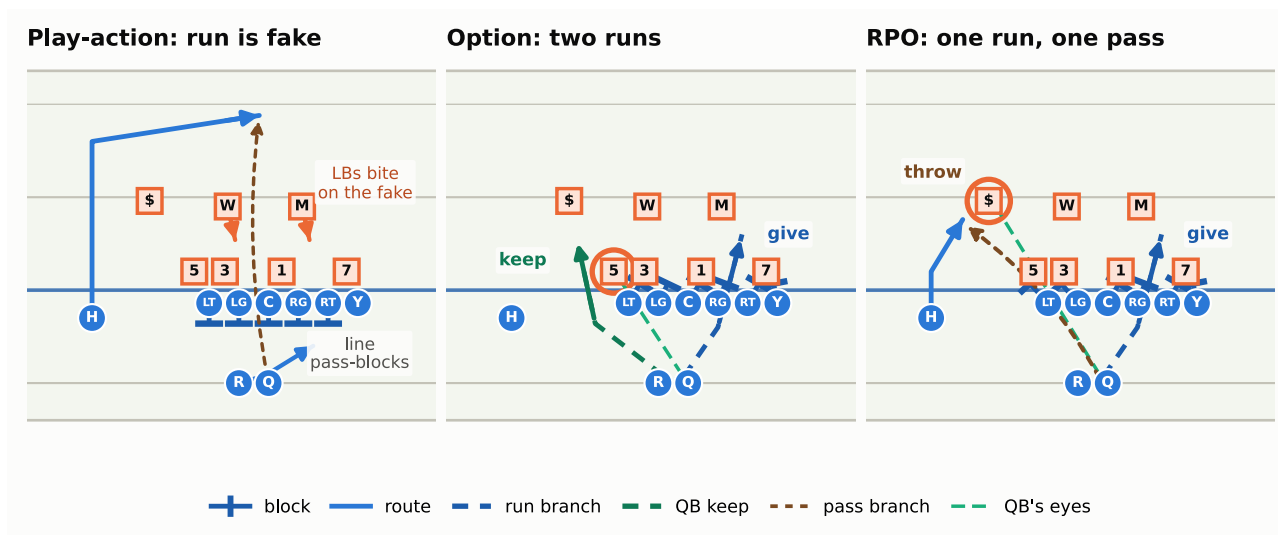


Figure 1: Same formation, same backfield action, three different plays. Left, play-action: the line pass-blocks (short bars a step behind the line), the back fakes a run without the ball, the linebackers step up to stop it, and the quarterback throws behind them to H; the pass was the plan from the start. Middle, the zone read (03-04): the line run-blocks to the right, nobody blocks the backside end (ringed), and the quarterback reads him to give or keep; both branches are runs. Right, an RPO: the line run-blocks to the right as in the middle (this time the left tackle blocks the end), but now the read is the nickel back (\$, ringed), who must help against the run and cover H; if he comes forward to the run, the quarterback throws to H in the space he left, and if he stays with H, the quarterback hands off.

Table 1: Three plays that start the same way.

	Play-action	Option (03-04)	RPO
What's real	Only the pass; the run is a fake	Both runs	Both the run and the pass
Who chooses, and when	The coordinator, before the snap	The read key's first step, after the snap	The quarterback, from a count before the snap or a read just after it
What the line does	Pass-blocks, made to look like run blocking	Run-blocks	Run-blocks, but in the NFL may not go more than a yard downfield if a pass is coming
Who is "left alone"	Nobody: the fake does the work	A defender in the run fit (usually the end)	A defender with a run job <i>and</i> a pass job
What the quarterback watches	Coverage, after the fake	The read key	The conflict defender, or the count

Read the table's middle column next to the right one. The **option** reads a defender whose only job is the run: the end can tackle the back or the quarterback, so the offense makes him pick one. The **RPO** reads a defender who has a run job and a pass job, and

makes him pick one of *those*. That is why it is called an option at all. It is the same arithmetic as the zone read, moved out of the backfield and into the space between the box and the receivers.

And read the left column against the right. **Play-action** also wants a linebacker to step up, but it needs him to *believe* the fake, and if he doesn't, the quarterback is holding the ball in a pocket with no plan B except a throw into coverage. The RPO doesn't need anyone to be fooled. If the linebacker steps up, throw. If he doesn't, the run is real, and the run is now going against a defender who is standing in the wrong place to stop it.

! Why it exists

RPOs exist because **modern defenses ask a few players to do two jobs at once, and an offense that waits to see which job they choose can always take the other one**. Spread formations pulled defenses out of the box, and defenses answered with hybrid players (the nickel back, the walked-down safety, the apexed linebacker) who stand halfway between the box and a slot receiver and promise to help with both. That promise is the target. The trade: an RPO commits the offense to a *simple* run and a *simple* throw (complicated plays don't fit in half a second), its linemen must obey a rule that run blocking normally ignores, and the quarterback has to make a clean decision while turning his body away from half the field. A wrong read is not a disaster, which is the beauty of it: the run was blocked anyway.

i Madden vs. real life

In the game, RPOs live in their own corner of the playbook, and you, the player, decide by pressing a button. That much is real: the quarterback really does decide. What the game hides is how *little* time he has and how *little* he is looking at. On a real post-snap RPO the quarterback is not scanning the field. He picks one defender before the snap, watches only him for about two-thirds of a second at the mesh, and either lets go of the ball into the back's belly or pulls it. Most of the time, he hands it off. An NFL RPO is mostly a run play with a safety valve, not a pass play that sometimes runs.

Who gets read: the conflict defender

Every RPO is built around one defender. Passing Fundamentals called him the **conflict defender**: a defender responsible for two things at once, so that whatever he does, he is wrong about one of them. On an RPO the two things are a run gap and a pass area.

The usual suspect is the defender you met in A First Look at Defense: the **overhang defender**, standing in an **apex** alignment halfway between the last man in the box and a slot receiver. On a run toward him, he has to come in and fill the edge of the box; on a pass, he has to widen and take the slot's short route. The apex is a promise to do both a step late, and the RPO calls the bluff. In a nickel defense he is usually the nickel back; against a single receiver, he may be a safety walked down or an outside linebacker.

He is the most common target, not the only one. Any defender whose run job and pass job pull in opposite directions will do:

- **An inside linebacker.** On a run, he steps forward to fill a gap; on a pass, he drops back into a zone (or turns to cover a back or tight end in man). Two steps forward for the run open the space behind him.
- **A safety who fits the run.** Against two-high shells, defenses often bring a safety down after the snap as the extra run defender. If he comes, the deep area he left is open.
- **Nobody in particular: the count.** On a pre-snap RPO, the quarterback reads no single man. He counts how many defenders are in the box and how many are out over the receivers, and goes where the numbers are.

What does the quarterback look for in a post-snap read? The same two things an option quarterback watches in a read key: **direction** and **eyes**. A defender who steps *forward* (or inside) with his eyes in the backfield has chosen the run. A defender who backpedals, widens, or turns his eyes to a receiver has chosen the pass. And the rule for doubt is the same as on most zone reads: **the run is the default**. The quarterback throws only when the defender has clearly taken himself out of the pass; if he's unsure, he hands off, because the run was blocked and a hesitant throw is the worst outcome on the field.

Pre-snap RPOs: count, then choose

The simplest RPO is decided before anyone moves. A **pre-snap RPO** gives the quarterback a run and a quick throw and tells him to pick one from the defense's alignment. No defender is read after the snap. The quarterback catches the ball and either hands it off or throws it, immediately, to the receiver the numbers favor.

The classic version is the **bubble RPO**: a run paired with a bubble screen to a slot receiver, the quick screen from The Quick Game in which the receiver bows back and out and catches the ball already running toward the sideline while the receiver outside him blocks. The quarterback counts twice:

1. **The box.** How many defenders are in the box, against how many blockers the offense has? With five linemen and a tight end, the offense has six.
2. **Outside.** On the bubble side, how many defenders are close enough to stop a bubble (count the ones within about seven yards of the line; a safety twelve yards deep arrives too late), against how many receivers the offense has there?

The defense has to put its players somewhere, so the two counts move against each other. Every defender who walks into the box to stop the run is one fewer outside, and every defender who stays outside over the slot is one fewer in the box.

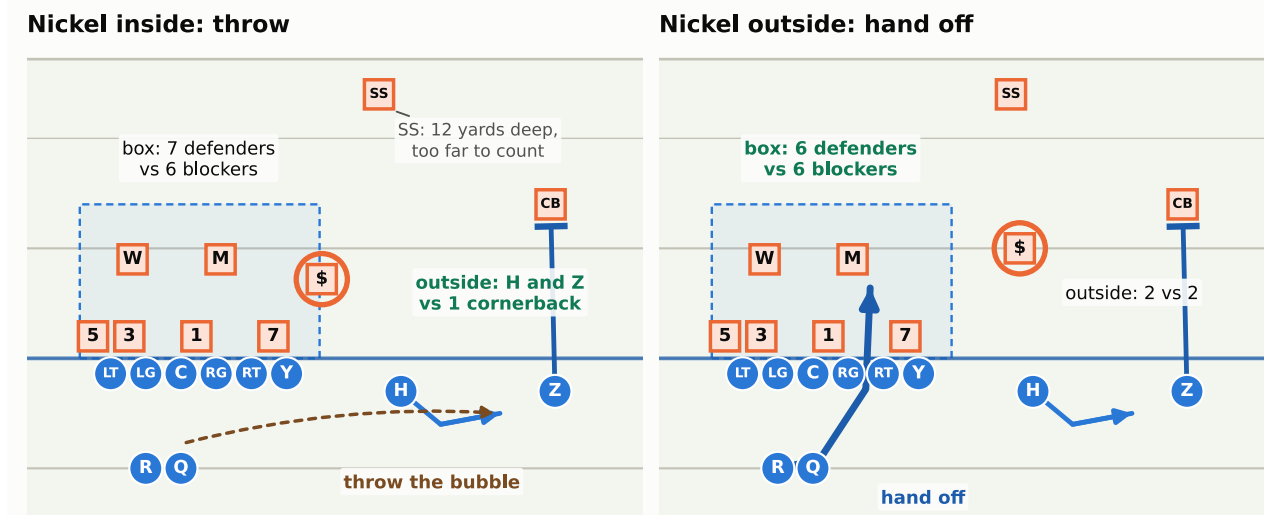


Figure 2: The bubble RPO decided by counting, against two alignments of the same nickel defense (the offense's right side; the box is shaded). Left: the nickel back (\$, ringed) has walked inside, next to the tight end. The box now holds seven defenders against six blockers, so a run is a man short; but outside, H and Z face only the cornerback, because the strong safety is twelve yards deep. Two against one: the quarterback throws the bubble to H at once, and Z blocks the cornerback. Right: the nickel back stays out over H. Outside is now two against two, with no advantage, but the box is six against six, a hat on every hat: the quarterback hands off, and H's bubble is just a decoy.

That's the whole logic, and it's worth saying why it's so hard to defend. The defense *wants* to put the nickel back in the box against a run-heavy offense: it's the extra man, the hat math of 02-05. And it *wants* to put him out over the slot against a passing offense. The bubble RPO says: pick one, and we'll run the other. The only escape is to have both, which means either bringing a safety down (and playing with one deep defender, which the offense's next play can attack) or moving after the snap so the count the quarterback made is wrong by the time the ball is out.

The offense pays a little, too. The bubble is a horizontal throw, so its ceiling is low; it gains yards only if the blocker outside wins and the receiver makes the first man miss. It is also a throw that, if it is dropped behind the line, becomes a backward pass and a live ball (The Quick Game has the rule). And it is predictable: an offense that throws the bubble every time the nickel walks in and then sprint back out at the snap, which a pre-snap count cannot see. That is the opening for the post-snap RPO.

Common misconception

"A pre-snap RPO is just an audible." Not quite. An **audible** changes the play: the quarterback calls out a new play at the line, and all eleven players have to hear it and change what they do. On a pre-snap RPO nobody changes anything. The line blocks the run, the back runs his track, the receivers run their routes or their screen, and only the quarterback's last action, hand off or throw, depends on the count. That's why RPOs are so fast to execute and so hard for a defense to read from the line: there is no new call to hear and no shuffle of feet when the check is made.

Post-snap RPOs: read a defender

A **post-snap RPO** moves the decision to after the snap. The line blocks the run, the back takes the ball into the mesh, and the quarterback rides him for a beat while he watches one defender. It is the zone read's mesh from 03-04, with the keep replaced by a throw. If the read defender plays the run, the quarterback pulls the ball and throws behind him. If he plays the pass, the quarterback lets go and the back runs.

The NFL's favorite version, and the flagship of this chapter, is the **glance RPO**. The run is inside zone (03-02). The pass is a *glance*: the receiver on the side away from the run takes a few hard steps upfield, then bends inside at a shallow angle, a skinny post that ends eight to twelve yards deep. The read is the linebacker on that side, the one who would fill the backside gaps against inside zone. His run job is to step forward; his pass job, in most zone coverages, is to drop toward the area the glance runs into.¹

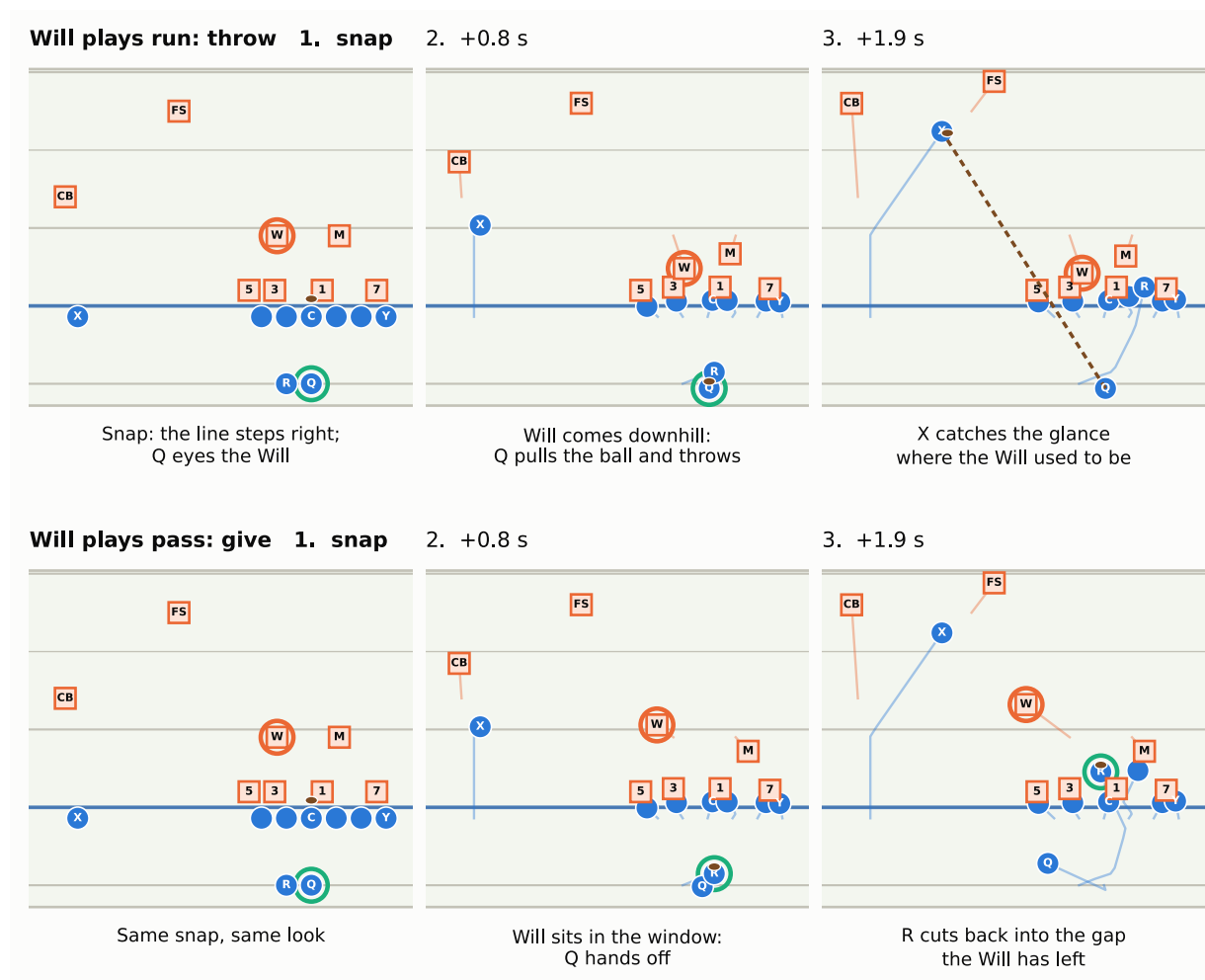


Figure 3: The glance RPO, two branches of one play (inside zone to the right; the quarterback reads the Will linebacker, ringed in orange; the green ring follows the ball). Top, the Will plays the run: at the mesh he is already two steps downhill into the backside gap, so the quarterback pulls the ball and throws the glance to X, who has bent inside behind him and in front of the deep safety; the back carries out his fake without the ball. Bottom, the Will plays the pass: he sits and drifts back toward the glance's window, so the quarterback hands off, inside zone runs against five box defenders with six blockers, and the back cuts back into the gap the Will has left. In both rows no lineman is more than a yard past the line when the decision is made. Simulated in Big Data Bowl coordinates; the trips side of the formation is outside the picture.

If the Will plays the run (top row of Figure 3), he has done what inside zone teaches him: see the line step, see the back coming, fill the gap. Against a plain handoff, that fill is how linebackers make tackles for no gain. Against the glance RPO it empties the zone behind him. The quarterback pulls the ball, the back carries out his fake (which still holds the other linebackers for a moment),

¹On the glance RPO's mechanics (a play that "incorporates a skinny post route into the RPO system"; if the read linebacker "drops into a zone, this is a hand-off," and if he "commits to the run," the receiver "replaces the linebacker with his route"), see "Purdue Football: Penn State on Film - RPO Glance Route," Hammer and Rails (Aug. 30, 2022): <https://www.hammerandrails.com/2022/8/30/23329647/purdue-football-penn-state-on-film-rpo-glance-route>; and Inside the Pylon / Scouting Academy, "ITP Glossary: Post-Snap RPO Reads": <https://scoutingacademy.com/itp-glossary-post-snap-rpo-reads/>. Which defender is read varies by team and coverage; reading the backside linebacker, as in this chapter's diagram, is one common version.

and the throw goes into the space between where the Will now is and where the safety is. The receiver catches it eight to twelve yards downfield, often running.

If the Will plays the pass (bottom row), he sits or sinks to take the glance away. Fine: now he isn't filling his gap, and inside zone is running against five box defenders with six blockers. Count the hats again. The defense started with six in the box against six blockers, an even count. The read took one of those six out of the run without anyone blocking him, so the right guard, who would normally have had to climb to the Will, is free to stay on the double team or climb to the Mike instead.

That last point is the quiet genius of the post-snap RPO, and it's the same as the option's: **the read is a block**. On plain inside zone, the backside linebacker is somebody's job, usually a guard who must leave a double team at just the right moment and travel four or five yards to find him. On the glance RPO, nobody has to go get the Will. Whatever he does, he has taken himself out of one of the two plays.

What can go wrong? Three things, and the defense will try all of them:

- **The defender lies.** A linebacker who takes one hard step forward and then drops sells "run" to the quarterback and still takes the glance away. Coaches call this playing *slow* or *mugging the window*; a quarterback reading a lie throws into a defender. The cure is the run-as-default rule: throw only on a defender who has clearly committed, two steps and shoulders turned, not one step.
- **Someone else takes the window.** If the Will fills the run and a safety rotates down into the space he left, the read was right and the throw is still covered. This is the RPO version of the scrape exchange (03-04), and it's coming in the defense section below.
- **The clock.** The decision has to be made at the mesh, inside about two-thirds of a second, and the ball has to be out right after, for a reason that is about rules rather than football. Hold that thought until *The one-yard rule*.

The slant RPO: reading the overhang

The glance reads a linebacker. The slant RPO reads the defender in the apex, and it is the purest picture of the conflict defender in this book.

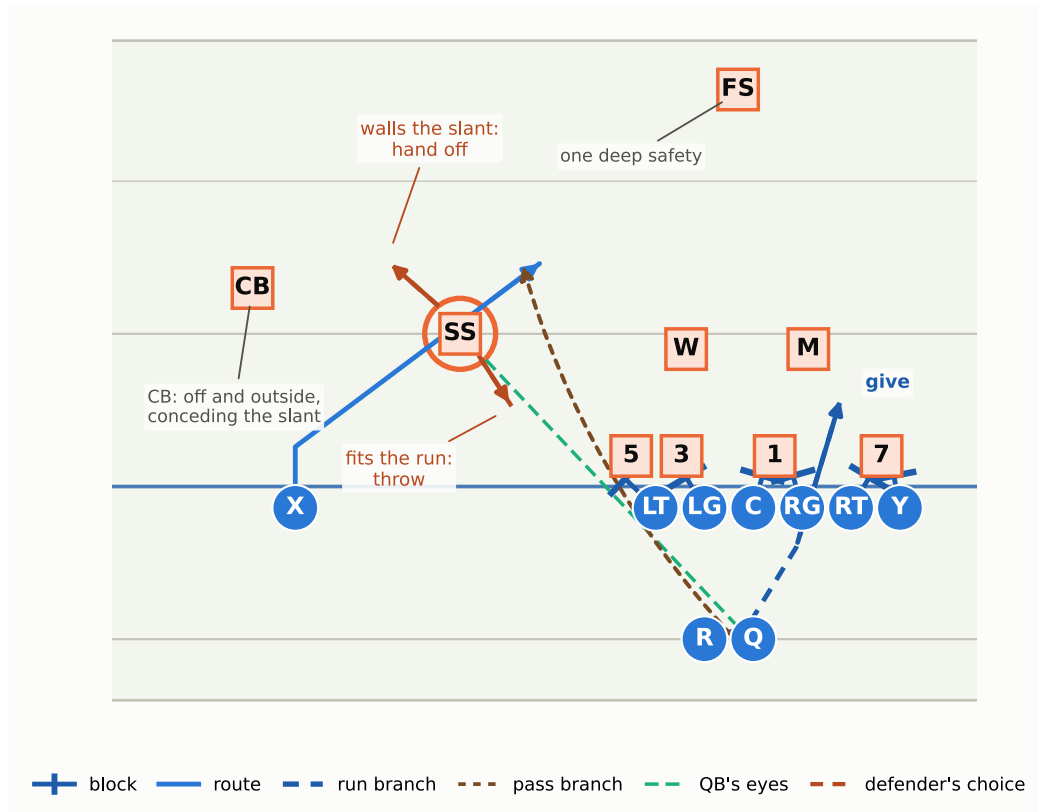


Figure 4: The slant RPO (the single-receiver side; inside zone goes away to the right). The defense is one-high, with the strong safety walked down as the overhang defender (ringed), halfway between the left tackle and X. X runs a quick slant whose path, five yards past the line, crosses almost exactly the spot where the safety stands. The quarterback reads him at the mesh. If the safety comes forward and inside to help against the run (upper red arrow), the slant arrives in the spot he has just left: throw. If he stays or widens to wall off the slant (lower red arrow), the slant is covered, but the backside of the run has lost its extra defender: hand off. Either way the cornerback, playing off and outside, is too far away to help on the slant.

The safety in Figure 4 is in the worst seat in football. The defense lined him up there because the offense's formation put three receivers on the other side, and somebody had to be the extra run defender on the weak side and also take away the quick throw to X. He is that somebody. Watch how the offense has arranged the field so that his two jobs point in opposite directions: the run's cutback lane is *inside* and *forward* of him, the slant's landing spot is *at* him, and he can't be in both places at the moment the ball is decided.

The cornerback matters too. The slant is a throw the corner can take away by lining up with **inside leverage** (on the receiver's inside shoulder, the term from *The Quick Game*). If he does, the quarterback's pre-snap look tells him the slant is dead, and good RPO offenses carry a second answer to that side (a bubble or a quick out, the **slant-bubble** and **slant-out** pairs) so the read still has two branches. This is where pre-snap and post-snap RPOs meet: the quarterback uses the pre-snap look to choose *which* pass is live and the post-snap read to choose *whether* to throw it.

Stick and the pop pass: two more throws behind the run

The same idea works with almost any quick throw that lands behind a run defender. Two you'll see every weekend:

- **The stick RPO.** Stick from *The Quick Game* puts a receiver at about six yards, settling in front of the quarterback, and another to the flat. As an RPO, the quarterback reads the defender over the slot (the nickel or the overhang): if he triggers on the run, the stick is open behind him; if he widens or sinks under the stick, hand off.
- **The pop pass.** A **pop pass** uses a player who looks like a run blocker. The tight end steps as if to block the end or a linebacker, then releases ("pops") straight up the field into the space behind linebackers who have stepped up to stop the run. The read is usually the linebacker over him. The pop is the RPO's cleverest trick, because linebackers are *taught* to read the tight end: a tight end who blocks means run. On the pop pass, the tight end shows them exactly that, for one beat, and they come.

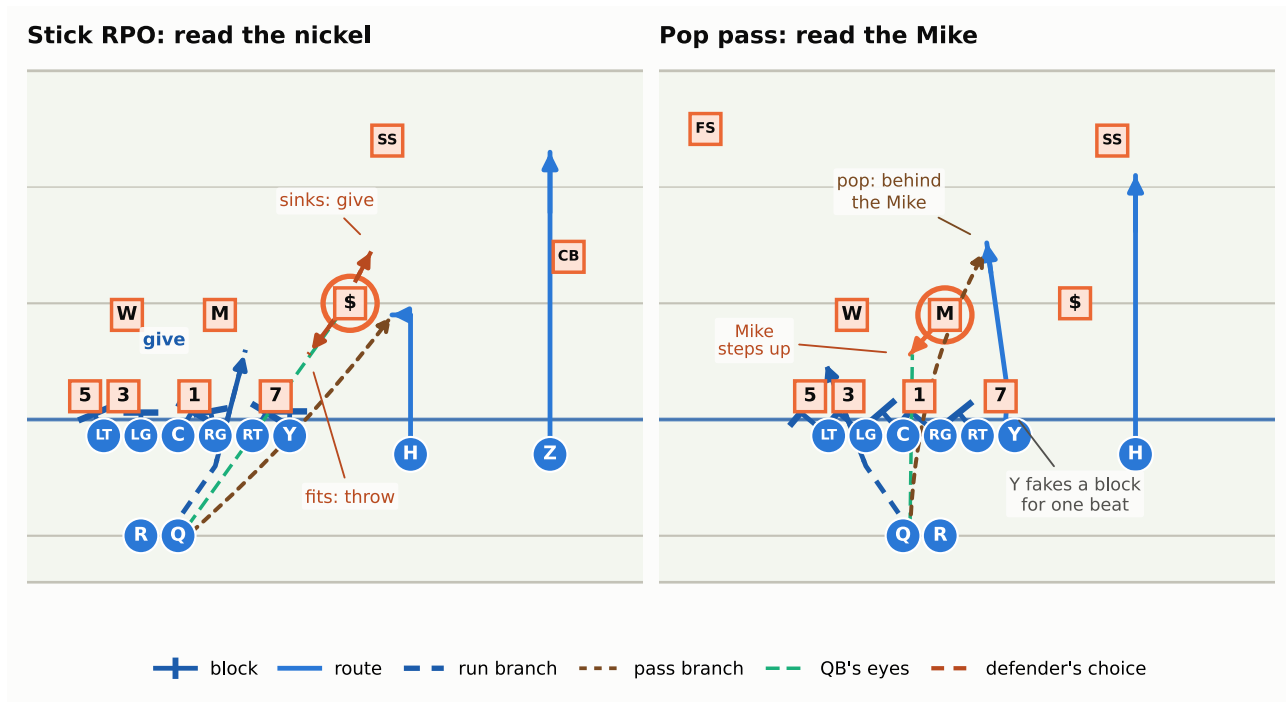


Figure 5: Two more pass branches behind an inside-zone run. Left, the stick RPO to the trips side: H runs a stick to six yards and Z clears the cornerback deep. The quarterback reads the nickel back (\$, ringed): if he steps up to fit the run coming toward him, he leaves the stick open; if he widens and sinks, hand off. Right, the pop pass, run going left: the tight end Y steps down as if blocking the 7-technique, then releases up the field behind the Mike linebacker (ringed), who reads Y's block as run and steps up into the hole; the quarterback pulls and throws to Y about seven yards downfield. Y is an eligible receiver, so his blocking fake can go anywhere; it's the five linemen who must stay near the line.

The pop pass also shows why RPOs and tight ends get along. A tight end is eligible, so his fake block can last as long as the play needs and his release can go anywhere. The five linemen can't do the same: they are *ineligible* receivers, and one rule about where they may be when a pass is thrown shapes every RPO in this chapter.

Three choices on one snap

Nothing stops an offense from putting the zone read and an RPO on the same snap. The quarterback reads two defenders in order, exactly like the triple option of 03-04, with a pass in the pitch man's role:

1. At the mesh, read the **backside end**: if he stays wide, give to the back; if he crashes on the back, pull the ball.
2. Now running, read the **overhang defender**: if he comes up to tackle the quarterback, throw the bubble or the slant past him; if he stays with the receiver, keep running.

College offenses called these *triple* RPOs or simply packaged plays, and they were common in the shotgun spread offenses of the 2000s and 2010s. They are rarer in the NFL, for two reasons you already know: the quarterback carries the ball into the second level on the keep, where he takes hits, and every extra beat of reading is a beat in which the linemen are run blocking further and further from the line.

The one-yard rule: why NFL RPOs look different

In football, only some players may catch a forward pass. The five interior linemen are **ineligible receivers** (The Twenty-Two), and the rules don't only stop them catching the ball, they stop them *running downfield before it is thrown*, because a lineman blocking a linebacker eight yards downfield while a receiver catches a pass behind him is a play the defense cannot fairly defend. The foul is called **illegal man downfield**, or **ineligible downfield** (the referee's announcement is often "ineligible man downfield").

Here is the NFL's version, as written in the 2026 rulebook:²

²NFL, 2026 Official Playing Rules of the National Football League, Rule 8, Section 3, Article 1 ("Ineligible Player Downfield": "On a scrimmage play during which a legal forward pass is thrown, it is a foul if the entire body of an ineligible offensive player ... is more than one yard beyond the line of scrimmage before the pass has been thrown"; Item 1, legally downfield after "initiating contact with an opponent within one yard of the line of scrimmage during his initial charge"; Item 2, illegally downfield; penalty "Loss of five yards from the previous spot"): <https://operations.nfl.com/rules-officiating/2026-nfl-rulebook>. The penalty summary lists the foul as "Ineligible player downfield on pass ... 8-3-1."

- On a play on which a legal forward pass is thrown, it is a foul if the **entire body** of an ineligible player is **more than one yard beyond the line of scrimmage** before the pass is thrown.
- **Exception:** a lineman who makes contact with a defender **within a yard of the line, on his initial charge**, may be driven or may drive his man further downfield *while still blocking him*. What he may not do is let go of that defender and keep moving forward, or go past a yard without having touched anyone.
- **Penalty:** five yards from the previous spot. A completed pass is wiped out; there is no loss of down.

Now set that against the run game. On inside zone, linemen double-team a defensive lineman and then one of them *climbs*: he lets go and runs to a linebacker four or five yards deep. That climb is the heart of zone blocking, and on a pass play it is exactly what the rule forbids: letting go of a defender inside one yard and then advancing.

College football draws the line somewhere else. Under the NCAA rule, no originally ineligible player may be **more than three yards beyond the neutral zone** (the strip the length of the ball between the two lines) until a legal forward pass **that crosses the neutral zone** has been thrown, also a five-yard penalty. (College judges *any part* of the body, the NFL the entire body.)³ Three yards is room for a guard to release toward a linebacker who has stepped up. Actually *blocking* him that far downfield before a pass that crosses the neutral zone is still offensive pass interference in college, unless the block began on the lineman's first charge within a yard of the neutral zone. And the "crosses the neutral zone" clause means a pass caught behind the line (a bubble, a now screen) doesn't trigger either rule at all. The NFL rule has no such clause; it applies to any legal forward pass.

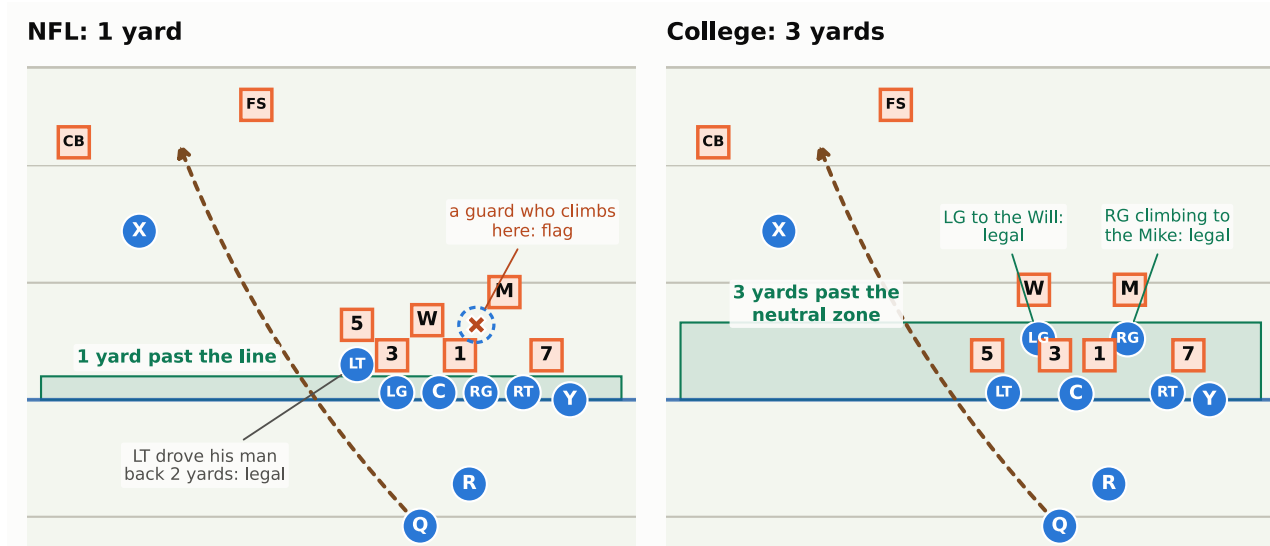


Figure 6: The same glance RPO at the moment the ball is thrown, under the two rulebooks (the left half of the line; the shaded strip is how far past the line of scrimmage an ineligible lineman may be when the pass leaves the quarterback's hand). Left, NFL: one yard. The linemen are still on the line, holding their first blocks. The right guard has to stay on the double team; a guard who had already climbed to the Mike (hollow circle) would be three yards downfield: flag. The left tackle, by contrast, engaged the end at the line and has driven him two yards back: legal, because he never let go. Right, college: three yards beyond the neutral zone. The guards have released toward both linebackers and are inside the three-yard limit when the ball leaves, so there is no foul, and the glance is thrown over the linebackers. (Had the guards already been blocking them that far downfield, college could call offensive pass interference instead.)

That difference is the single biggest reason an NFL RPO looks different from a college one:

- **NFL RPO throws are quick, and most are short.** The decision is made at the mesh and the ball is out within about a second of the snap, before any lineman has had time to drift past a yard. Many of the throws, the bubble and the now screen, land at or behind the line. (The data below shows how strong this is.)

³NCAA Football Rules (2026 edition), Rule 7-3-10: "No originally ineligible receiver shall be or have been more than three yards beyond the neutral zone until a passer throws a legal forward pass that crosses the neutral zone. A player is in violation of this rule if any part of their body is beyond the three-yard limit"; penalty five yards from the previous spot. Rule 7-3-8-b: offensive pass interference is "contact by a Team A player beyond the neutral zone that interferes with a Team B player during a legal forward pass play in which the forward pass crosses the neutral zone," except when an ineligible player "immediately charges and contacts an opponent at a point not more than one yard beyond the neutral zone and maintains the contact for no more than three yards beyond the neutral zone" (in 2026 its penalty became 10 yards). Text as reproduced in the 2026 NCAA rules and interpretations study compilation posted by the Amarillo Football Officials Association: <https://www.amarefs.org/wp-content/uploads/2026/07/RB-Combined-2026-1.pdf>; Rule 7-3-10 also at the San Diego County Football Officials Association: <https://www.sdcfoa.org/ncaa/rule-7-snapping-and-passing-the-ball/college-7-3-10>.

- **NFL RPO blocking stays at the line.** Instead of climbing, linemen hold their first blocks, or the offense uses blocking that naturally stays on the line (down blocks, double teams that don't release) and lets the read handle the linebacker. The glance RPO is popular in the NFL for exactly this reason: the read takes care of the man the climbing guard would have blocked.
- **College RPOs can be “run first, throw later.”** With three yards and no limit at all on throws behind the line, college linemen can run-block for longer and drift up to three yards downfield, the quarterback can ride the mesh longer, and the pass can go farther downfield: seams and posts over linebackers who have stepped up to the run.

Why hasn't college changed? It nearly did. In 2015 the NCAA's rules committee proposed cutting its limit to one yard, and the oversight panel tabled the proposal; the rule is still three yards for 2026.⁴ In the NFL the one-yard rule is not new, but RPOs put it under strain on every snap: an RPO whose throw comes out a beat late is exactly the play on which a run-blocking lineman has drifted too far. You will see in the numbers that RPOs draw far more than their share of the flags.

i Go deeper

Receivers who block too early. The one-yard limit is about ineligible linemen, but a cousin rule binds receivers. In the current NFL rulebook, blocking a defender more than a yard past the line *before* a pass is thrown is offensive pass interference, and pass interference can occur on a pass thrown from behind the line “regardless of whether it crosses the line.”⁵ On a bubble RPO, then, the receiver blocking the cornerback wants his block to land as the ball leaves the quarterback's hand, not before, which is one more reason the bubble comes out at once. Whether a block came “clearly prior” to the throw, the rule's words, is a judgment made at full speed; Bill Belichick, asked about RPOs in 2019, called the downfield-blocking question “a tough call for the officials.”⁶

Where RPOs came from

Nobody invented the RPO in one moment, and several people say they did. What is not disputed is the ingredients: the option's habit of reading a defender instead of blocking him, and the spread offense's habit of giving the quarterback a quick throw to whichever side the defense left light. Put a quick throw on a running play and you have it.

- **Air Raid “tags” and the 1990s.** Coaches in Hal Mumme and Mike Leach's Air Raid tradition, at Kentucky in the late 1990s and Leach's Oklahoma in 1999, tagged quick screens onto runs so the quarterback could throw if the defense overloaded the box. Rich Rodriguez, whose zone read grew out of a fumbled practice handoff at Glenville State (03-04), built quick slants and bubbles into his zone-read offense at Tulane and Clemson.⁷
- **Claims to the invention.** Urban Meyer says his Utah teams found it by accident in the early 2000s, when a receiver missed his block and ran a bubble instead, the quarterback threw to him, and assistants Dan Mullen and Kyle Whittingham wrote it into the playbook.⁸ Brett Favre has said that, bored with handing off, he began telling receivers to run quick routes on running plays and threw to them when the look was right, before he had told his coach.⁹ Treat all of these as what they are: stories coaches tell about a thing that was in the air.

⁴SI Wire, “NCAA tables proposed ineligible downfield rule” (Mar. 6, 2015), on the Playing Rules Oversight Panel postponing a move from three yards to one: <https://www.si.com/college/2015/03/06/ncaa-ineligible-downfield-rule>; FootballScoop, “The ineligible receiver downfield rule has been tabled by the NCAA”: <https://www.footballscoop.com/2025/03/04/ineligible-receiver-downfield-rule-tabled-ncaa>. The 2026 NCAA rule still reads three yards (Rule 7-3-10; see the previous note).

⁵NFL, *2026 Official Playing Rules*, Rule 8, Section 5, Article 1 (pass interference “can only occur when a forward pass is thrown from behind the line of scrimmage, regardless of whether the pass is legal or illegal, or whether it crosses the line”) and Article 4(a) (offensive pass interference includes “blocking more than one yard beyond the line of scrimmage by an offensive player clearly prior to a pass being thrown”): <https://operations.nfl.com/rules-officiating/2026-nfl-rulebook>.

⁶Darren Hartwell, “Is Bill Belichick sending a message to NFL refs with this RPO comment?,” NBC Sports Boston (Dec. 6, 2019): “The only problem (with) the RPO play in general is just the offensive pass interference: blocking downfield conflicting with the pass. ... It's a tough call for the officials”: <https://www.nbcsportsboston.com/nfl/new-england-patriots/is-bill-belichick-sending-a-message-to-nfl-refs-with-this-rpo-comment/379873/>.

⁷Dennis Dodd, “How the RPO has brought the NFL and college football together in an offensive revolution,” CBS Sports (Nov. 8, 2018), on Rich Rodriguez as “one of the founding fathers of the zone read offense that birthed RPOs” and Mike Leach bringing his Air Raid offense to Oklahoma in 1999: <https://www.cbssports.com/college-football/news/how-the-rpo-has-brought-the-nfl-and-college-football-together-in-an-offensive-revolution>; Mike Calabrese, “EA Sports College Football 25: RPOs & Other Offensive Innovations,” Action Network (June 2024), on Mumme's Kentucky giving Tim Couch “bubble screens or handoffs to the running back” and Rodriguez at Tulane and Clemson balancing “slants, bubble screens and short hitches” with read options: <https://www.actionnetwork.com/ncaaf/ea-sports-college-football-25-rpos-offensive-innovations>.

⁸Keith Farner, “WATCH: Urban Meyer takes credit for inventing the RPO, explains how it happened,” Saturday Down South (Sept. 3, 2019): <https://www.saturdaydownsouth.com/sec-football/watch-urban-meyer-takes-credit-for-inventing-the-rpo-explains-how-it-happened/> (“it was a missed assignment by a wide receiver”). Meyer coached Utah in 2003 and 2004.

⁹Jared Dubin, “Here's Brett Favre explaining how he basically invented RPOs because he was bored,” CBS Sports (June 11, 2018), from Favre's appearance on Jon Gruden's QB Camp: <https://www.cbssports.com/nfl/news/heres-brett-favre-explaining-how-he-basically-invented-rpos-because-he-was-bored>.

- **The college spread, 2000s–2010s.** High-school and college spread offenses made packaged plays routine. When Robert Griffin III arrived in Washington in 2012 from Art Briles’s Baylor, Mike Shanahan borrowed Baylor’s packaged plays, run blocking up front and pass routes outside, with Griffin choosing after seeing the defense. After Griffin’s first game, Shanahan put it this way: “we had running plays called and all of a sudden they gave us certain looks to take the running game away, and we threw a couple of bubbles early.”¹⁰ The full lineage, from the Wing-T and the wishbone through the spread option, is in *The College Laboratory*.
- **The NFL boom, 2016–2018.** The tree that made RPOs a fixture of NFL offense is Andy Reid’s. By Pro Football Focus’s charting, the share of NFL plays that were RPOs roughly doubled from about 4% in 2016 to about 8% in 2018; Reid’s Chiefs led the league in RPO usage in 2017 (17.1% of their plays) and 2018 (25.0%), and Doug Pederson, who had been Reid’s offensive coordinator in Kansas City from 2013 to 2015, ranked second with the Eagles in 2017 and won a Super Bowl with them (see the film room). Matt Nagy, Pederson’s successor in Kansas City, took the idea to Chicago, where the Bears went from 2.7% RPOs in 2017 to 19.2% in his first season, 2018.¹¹
- **Since then.** PFF’s league-wide rate kept climbing to a peak of about 10.5% of plays in 2023, then fell to about 9.0% in 2024 and 8.0% in 2025.¹² RPOs stopped being a trend and became furniture: almost every NFL offense has a handful in every game plan.

i Film room: Philadelphia Eagles 2017

The 2017 Eagles are the RPO’s breakthrough on the NFL’s biggest stage. Doug Pederson built RPOs into the offense for Carson Wentz, and when Wentz tore his ACL against the Rams in December, he leaned on them even harder for the backup, Nick Foles. The logic was the RPO’s sales pitch in miniature: Foles was a less instinctive improviser than Wentz, so the coaches gave him throws whose answer the defense handed him at the snap. “It’s centered around accurate throwing, good decision-making and good execution,” offensive coordinator Frank Reich said in January 2018. “When we use it, Nick has shown a great aptitude of doing that very well.” By PFF’s count, Foles completed 93.9% of his RPO throws that season. He completed 26 of 33 passes for 352 yards and three touchdowns in the 38–7 NFC Championship win over Minnesota, and two weeks later, on February 4, 2018, in Minneapolis, he threw for 373 yards and three touchdowns (and caught one) as the Eagles beat New England 41–33 in Super Bowl LII. He was the game’s MVP.¹³

What to notice on the Eagles’ playoff film: how often the ball came out of Foles’s hand almost at the mesh, on quick slants, glances and bubbles, before the linebackers had finished stepping toward the run fake; and how the line’s first step was identical on the handoffs and the throws. Defenses that sat back to take away the throws got Jay Ajayi, LeGarrette Blount and Corey Clement running against light boxes.

¹⁰Gregg Rosenthal, “Mike Shanahan implements Baylor plays to help RG3,” NFL.com (Sept. 12, 2012), quoting Shanahan and citing Chris B. Brown’s Grantland analysis of “packaged plays”: <https://www.nfl.com/news/mike-shanahan-implements-baylor-plays-to-help-rg3-0ap1000000061442>.

¹¹Anthony Treash, “The RPO and the Andy Reid coaching tree,” PFF (May 17, 2019): <https://www.pff.com/news/pro-the-rpo-and-the-andy-reid-coaching-tree> (league RPO rate from “4% of plays in 2016 to over 8% of plays last year”; the Chiefs led in 2017 at 17.1% and 2018 at 25.0%; the Eagles ranked eighth in 2016, second in 2017 and third in 2018; Pederson “served on Reid’s staff in Kansas City from 2013 to 2015”; the Bears went from 2.7% in 2017 to 19.2% in 2018).

¹²Daire Carragher, “Analyzing the effectiveness of RPOs over the last five NFL seasons,” PFF (June 25, 2026): <https://www.pff.com/news/nfl-analyzing-the-effectiveness-of-rpos-over-the-last-five-nfl-seasons> (league RPO rate by season: 3.84% in 2016, 6.14% in 2017, 7.85% in 2018, 7.63% in 2019, 8.66% in 2020, 9.86% in 2021, 9.97% in 2022, 10.52% in 2023, 8.96% in 2024, 7.95% in 2025).

¹³Tim McManus, “Eagles pull from Chip Kelly playbook to awaken 2013 Nick Foles,” ESPN (Jan. 26, 2018): https://www.espn.com/blog/philadelphia-eagles/post/_id/24097/eagles-pull-from-chip-kelly-playbook-to-awaken-2013-nick-foles (Reich’s quote); Jack Browne, “Blueprint for an upset: How the Eagles can shock the Patriots,” theScore (before Super Bowl LII, 2018), citing PFF for Foles’s 93.9% completion rate on RPOs “this season”: <https://www.thescore.com/nfl/news/1474361>; “Eagles quarterback Nick Foles wins Super Bowl LII MVP,” NFL.com (Feb. 4, 2018): <https://www.nfl.com/news/eagles-quarterback-nick-foles-wins-super-bowl-ii-mvp-0ap30000000914463>; “Super Bowl LII,” Wikipedia: https://en.wikipedia.org/wiki/Super_Bowl_LII (U.S. Bank Stadium, Minneapolis). Wentz’s injury: Associated Press, “AP sources: Doctors believe Eagles’ Carson Wentz tore ACL, out for year,” via Boston.com (Dec. 11, 2017), on the Dec. 10 win at the Los Angeles Rams: <https://www.boston.com/sports/nfl/2017/12/11/ap-sources-doctors-believe-eagles-carson-wentz-tore-acl-out-for-year>. Game lines (NFC Championship, Jan. 21, 2018: Foles 26 of 33, 352 yards, 3 TD, Eagles 38–7; Super Bowl LII: 373 yards, 3 TD passes, a TD catch from Trey Burton, Eagles 41–33): nflverse play-by-play, this book’s check.

i Film room: Kansas City Chiefs, 2017 to 2025

Andy Reid's Chiefs have used RPOs more, and for longer, than any other NFL team: first in the league in PFF's count in 2017 and 2018, and first again in 2025 in FTN's charting (see the next section). The quarterback changed from Alex Smith to Patrick Mahomes; the idea didn't. Reid's RPOs are built for his receivers' speed, glances and slants behind linebackers, quick screens to the fastest player on the field, and since 2020 Mahomes has thrown more passes on RPOs than any other NFL quarterback.¹⁴ The full Reid story is in the Chiefs case study.

What to notice: on a Chiefs first down from the gun, find the defender over the slot or the backside linebacker before the snap, then watch Mahomes's head at the mesh. He is not looking at the back. When the ball comes out, it comes out in about a second, and it goes to the spot that defender just left.

i Film room: Jalen Hurts's Eagles, 2022

The RPO came back to Philadelphia in a different form with Jalen Hurts, whose running made every read harder to defend. In 2022, the season the Eagles reached Super Bowl LVII, FTN's charting has them using RPOs on a larger share of their plays than any other team. Their offensive coordinator that year, Shane Steichen, became the Colts' head coach in 2023, and the 2023 Colts used RPOs on a bigger share of their plays than anyone in the four seasons of FTN data: a scheme travelling with its coach.¹⁵ The Eagles' RPOs, the run game and the tush push are the subject of their own case study.

What to notice: how Hurts's legs change the read. With a quarterback who can also keep on a zone read, the backside end and the overhang defender both have to respect him, so a single snap can carry a give, a keep and a throw.

What the numbers say

The NFL's play-by-play doesn't say whether a play was an RPO: a handoff is a handoff and a pass is a pass. But FTN's charters, who watch every NFL play, flag RPOs, and the flag is in the nflverse data from 2022 on. The numbers come with a caveat that is worth learning once: **charting services disagree**. PFF's rates for the same seasons run well above FTN's (about 8 to 10.5 percent of plays against FTN's 4 to 6), because deciding from film whether a handoff "was" an RPO (did a receiver run a real route? was the quarterback reading someone?) is a judgment, and charters judge differently. Use one series at a time, and compare teams within it.

¹⁴Same PFF article (Carragher, 2026): Mahomes led all quarterbacks in RPO pass attempts since 2020 (391), ahead of Aaron Rodgers (287) and Josh Allen (268).

¹⁵FTN charting via nflverse, this chapter's calculation (see the data figure): highest team RPO share, 2022 Philadelphia; 2023 Indianapolis. Steichen was the Eagles' offensive coordinator in 2021 and 2022 and was hired as the Colts' head coach in February 2023: "Shane Steichen," Wikipedia: https://en.wikipedia.org/wiki/Shane_Steichen.

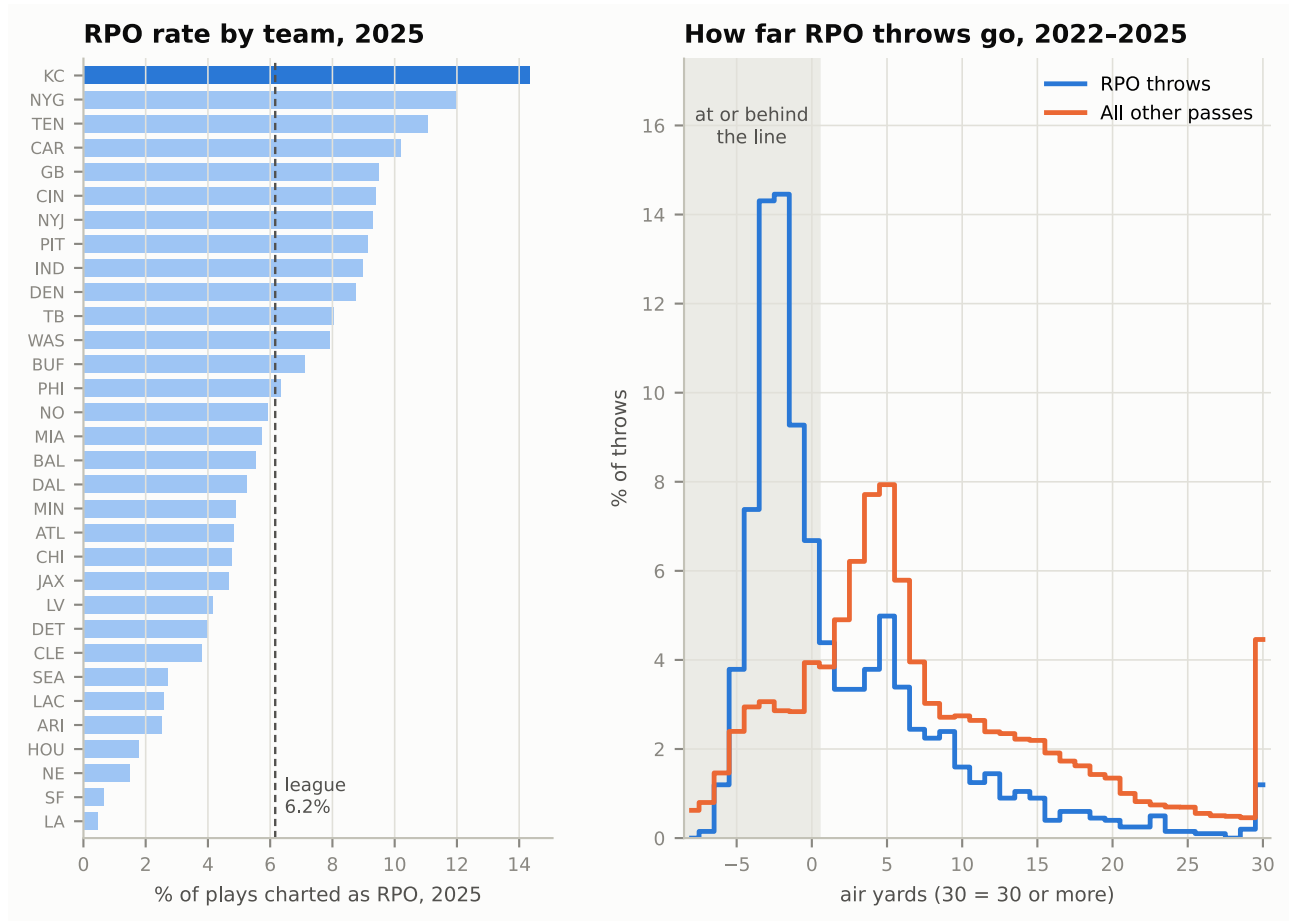


Figure 7: RPOs in FTN's NFL charting. Left: the share of each team's offensive plays charted as RPOs, 2025 regular season (runs and passes, scrambles included, kneel-downs and spikes excluded). Right: how far past the line RPO throws travel in the air, against all other passes, 2022–2025 regular seasons; air yards at or behind zero (shaded) are thrown to a receiver at or behind the line of scrimmage. RPO throws pile up behind the line, the bubbles and now screens that the NFL's rules favor; other passes spread far downfield. Data: nflverse play-by-play joined to FTN charting (load_ftn_charting) via nflreadpy; FTN's RPO flag is a charter's judgment, and other services count more RPOs.

Five things to read from Figure 7 and the numbers behind it.

RPOs are a small, steady slice of NFL offense. FTN charted 5.0% of plays as RPOs in 2022, 5.3% in 2023, 3.6% in 2024 and 6.2% in 2025 (the 2024 dip is larger than PFF's and may partly be charting). The leaders are the teams this chapter has already met: PHI in 2022 (13.5%), IND in 2023 (18.3%), and Kansas City in 2025 (14.4%), more than twice the league rate. At the other end, several teams barely use them, a reminder that the RPO is a scheme choice, not a law of modern football: a team that runs mostly from under center with wide-zone blocking, as the Shanahan-tree offenses do, has little use for a play whose throws must come out of a shotgun mesh in a second.

Most RPOs are runs. On 69% of the charted RPOs from 2022 to 2025 the ball was handed off (79% in 2025). That is the “run with a safety valve” idea from the Madden callout, in data. It also means the RPO's main effect is invisible in a box score: its product is the defender who wasn't in the run fit.

NFL RPO throws are short, and very often behind the line. 57% of RPO throws were caught (or aimed) at or behind the line of scrimmage, against 21% of other passes; only 11% travelled more than ten yards in the air, against 30% of other passes. That shape is the one-yard rule at work. A throw behind the line lets the linemen keep run blocking for an extra beat without risk of the five-yard flag hanging over a downfield throw, and a throw beyond the line has to leave so fast that it can't be far beyond it.

The rule bites. From 2022 to 2025 the NFL called 246 ineligible-downfield fouls on passes in the regular season. 24% of them came on plays FTN charted as RPOs, though RPOs were only 5.0% of plays: per snap, an RPO is many times more likely to draw the flag than any other play.

Per play, the throws look great and the runs look bad, and neither number means what it seems. RPO throws averaged +0.06 EPA with a 79% completion rate; RPO handoffs averaged -0.05. Taken together, RPOs come out close to ordinary plays (-0.019

EPA per play against -0.005 , and a success rate of 42.8% against 43.5%). The split is *selection*, exactly as with the quarterback keeps of 03-04: the quarterback throws precisely when the throw is open and hands off when it isn't, so the throws are the good half of the reads by construction. And notice what the box count does, weakly: when FTN counted six defenders in the box, 28% of RPOs were thrown; with seven, 33%; with eight or more, 40%. More defenders near the ball, more throws: the counting logic of the bubble RPO, visible in charting that only sees the alignment at the snap and not the read after it.¹⁶

PFF's charting, with its larger RPO count, frames the trade a slightly different way: from 2021 to 2025, NFL RPOs gained fewer yards per play than other plays (4.53 against 5.43) but succeeded more often (52.3% against 48.1%).¹⁷ Both series agree on the shape. RPOs are not big-play machines; they are **staying-on-schedule** machines, a way to make first-and-10 into second-and-5 without asking anyone to win a hard block.

How defenses answer

Defenses have had a decade to study NFL RPOs, and the answers all attack one of the RPO's three parts: the read, the throw, or the run behind it. This is a preview; the defense's side gets its full treatment in Run Fits: Gap Integrity, Force, Spill, and Box Math.

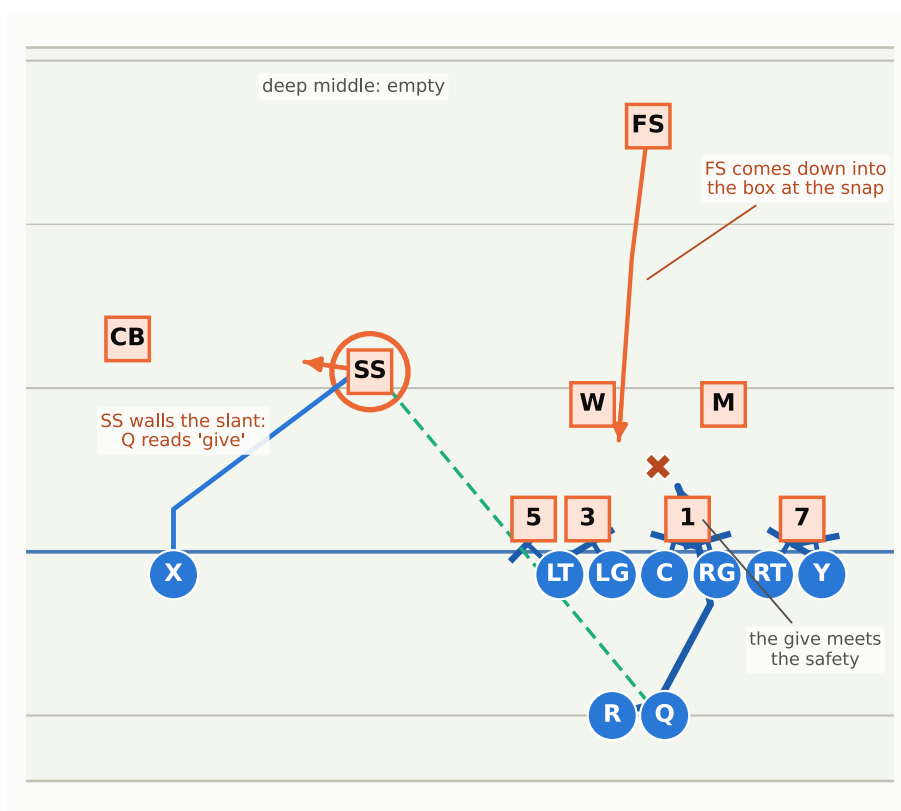


Figure 8: Making the read lie: one answer to the slant RPO of the earlier figure. The overhang safety (ringed) is told to play the slant no matter what, and he widens at the snap, so the quarterback reads 'give.' But the free safety, deep in the middle, has a run job on this call: he comes downhill at the snap into the backside of the box, the spot the overhang defender would have filled. The read was right and the run still meets an extra defender. The price: with the free safety in the box, nobody is left in the deep middle.

- **Make the read lie.** Give the read defender a rule (always play the pass) and put someone else in his run fit, usually a safety rotating down after the snap, as in Figure 8. It is the scrape exchange of 03-04 with a safety instead of a linebacker, and it's built on the post-snap rotations of Disguise and Rotation. The cost is the deep defender who left.
- **Play two-high and fit the run from depth.** Two-high shells, especially the quarters family of 06-05, keep the overhang defender honest on the pass and let a safety who starts ten yards deep become the extra run defender once the ball is handed

¹⁶This book's calculation from nflverse play-by-play (`load_pbp`) joined on game and play ID to FTN charting (`load_ftn_charting`, flag `is_rpo`), regular seasons 2022–2025, plays with `play_type` pass or run and a down (so scrambles and sacks on RPO looks are included; spikes and kneel-downs are not). Air yards: `pbp air_yards` on RPO passes. Box counts: FTN `n_defense_box`. Ineligible-downfield fouls: `pbp penalty_type == "Ineligible Downfield Pass"` on any regular-season play, joined to FTN's flag. EPA and success as defined in Expected Points, EPA and Success Rate.

¹⁷Same PFF article (Carragher, 2026): 2021–2025, RPOs 4.53 yards per play and a 52.3% successful-play rate, non-RPO plays 5.43 and 48.1%. PFF's success definition is its own and differs from the EPA-based success rate used elsewhere in this book.

off. Against a pre-snap count, this is a defense that shows six in the box and plays like seven. Its weakness is the run itself: a safety arriving from ten yards deep arrives late.

- **Take away the conflict with man coverage.** In man coverage, the defender over the slot isn't torn between a gap and an area: he has a man, and he goes with him. Nobody is in conflict, so there's nothing to read. The price is the one man coverage always pays: defenders with their backs to the ball, and a box whose count the run can attack.
- **Squeeze the clock.** Hold the read defender's decision as long as possible (a slow-play, a late trigger) so the quarterback rides the mesh a beat longer, and the linemen, run blocking, drift past a yard. The flag is a defensive weapon too, and a defense that has watched the tape knows which linemen climb early.
- **Hit the quarterback.** Every RPO exposes him for a moment with his eyes on one defender and his body turned toward the back; pressures that come free off the edge, and defenders who hit him after the throw within the rules, make him hurry the read.

Offenses answer back, as always: with motion that changes who the conflict defender is just before the snap, with RPOs that read the *safety* instead of the overhang, and with play-action that looks exactly like the RPO for the first second and then throws deep over the safety who has learned to come down. That conversation, how one play makes the next one work, is the subject of Constraint Theory, where the RPO appears as a **constraint play**: a play that punishes a defense for overplaying the offense's base run.

Watch for it

On Sunday: "Is he reading you?" Every RPO has a defender being read. Find him before the snap and you'll see the decision before the replay does.

Before the snap

- **Shotgun or pistol, back beside the quarterback, receivers spread.** That's the RPO look. Not every snap from it is an RPO, so treat it as "possible."
- **Count the box, then count outside.** Six or seven in the box? How many defenders are within seven yards of the line over the slot receivers? If the box is heavy and the slots are light, expect a throw (a bubble or now screen) the instant the ball is snapped.
- **Find the apex.** Who stands halfway between the box and a slot receiver? On the single-receiver side, is a safety walked down? That's the likely read. Say it out loud: "he's being read."

At the snap

- **Line run-blocking, quarterback's head turned away from the back.** On a plain handoff, the quarterback looks at the back. On a post-snap RPO, at the mesh, **his eyes are on a defender.** Find the defender he's looking at.
- **Watch that defender's first two steps, not the ball.** Forward and inside: predict *throw*, behind him. Backward, widening, or eyes on a receiver: predict *give*.
- **Count to one.** If the ball isn't out of the quarterback's hand by about a second, it's probably a handoff.

After the play

- **On the replay, freeze the frame when the ball leaves the quarterback's hand and look at the linemen.** Is anyone more than a yard past the line without a defender in his hands? That's the flag the umpire is looking for. Did one get thrown?
- **On the next snap,** did the read defender do the opposite? Defenses adjust within a drive.

For your prediction log (01-06): on every RPO look, write the defender you think is being read and your call (G or T) from his first steps; afterward, write what happened.

Predict the play

💡 Predict the play

Drill 1: count both sides. First-and-10 at midfield. The offense is in the shotgun, 11 personnel, trips to the right with the tight end attached, the back offset left. It has thrown the bubble RPO twice already today. This time the defense shows the picture below before the snap. Count the box and count outside the box on the trips side. Hand off or throw the bubble? And is there anything on the left side the quarterback should like?

Drill 1: 1st and 10, the trips side crowded

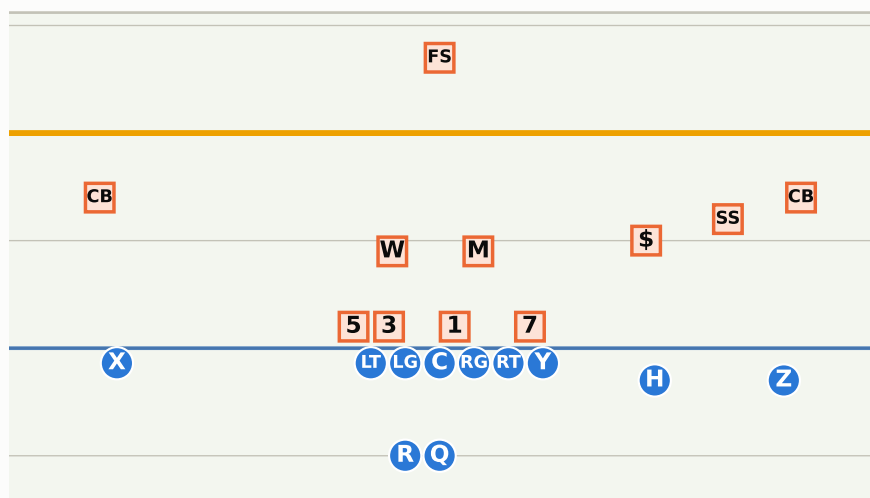


Figure 9: Drill 1: first-and-10, gun 11 personnel, trips right with the tight end attached. The defense has the nickel back over H and the strong safety walked down outside him, six yards deep; the free safety is alone in the deep middle. On the left, X is alone against the cornerback, seven yards off him.

Answer 1 is at the end of the chapter. Try it first.

💡 Predict the play

Drill 2: freeze frame. Second-and-8. The offense runs the slant RPO of Figure 4 to the left, the inside-zone run going right. The picture below is frozen about six tenths of a second after the snap, with the ball in the back's belly at the mesh. The overhang safety (ringed) has taken two quick steps. Does the quarterback give or throw? What happens if the cornerback had lined up on X's inside shoulder instead?

Drill 2: 2nd and 8, frozen at the mesh

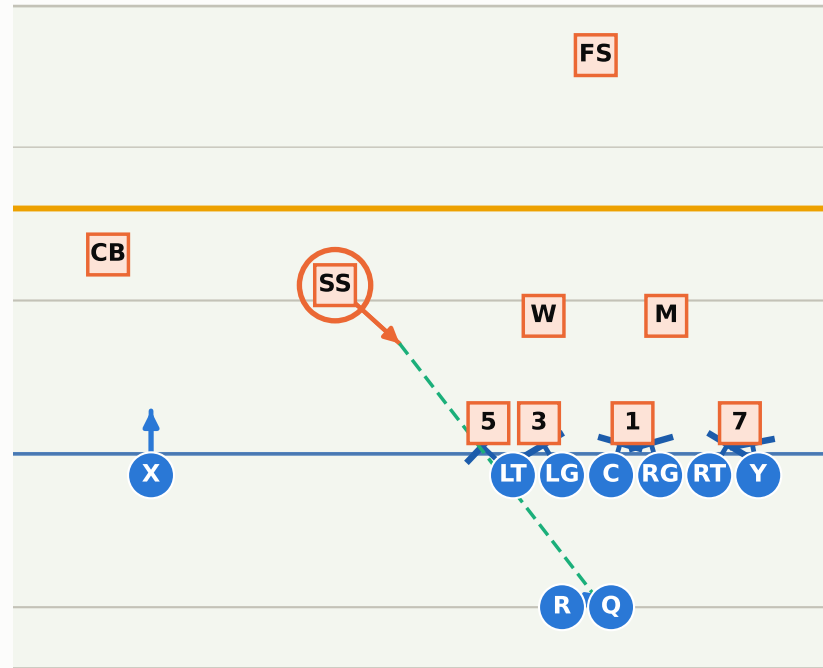


Figure 10: Drill 2: the slant RPO frozen at the mesh, about 0.6 seconds after the snap. Lines show where each player has moved since the snap: the line run-blocking right, X two steps into his stem, and the overhang safety's first two steps.

Answer 2 is at the end of the chapter. Try it first.

💡 Predict the play

Drill 3: flag or no flag? A glance RPO goes for 15 yards. On the replay, frozen at the instant the ball left the quarterback's hand, you see the picture below. The left guard double-teamed the 3-technique with the left tackle, let go, and is now three yards past the line on his way to the Will. The right tackle engaged the 7-technique at the line and has driven him two yards backward, still locked on. Is there a foul in an NFL game? In a college game?

Drill 3: the ball is out

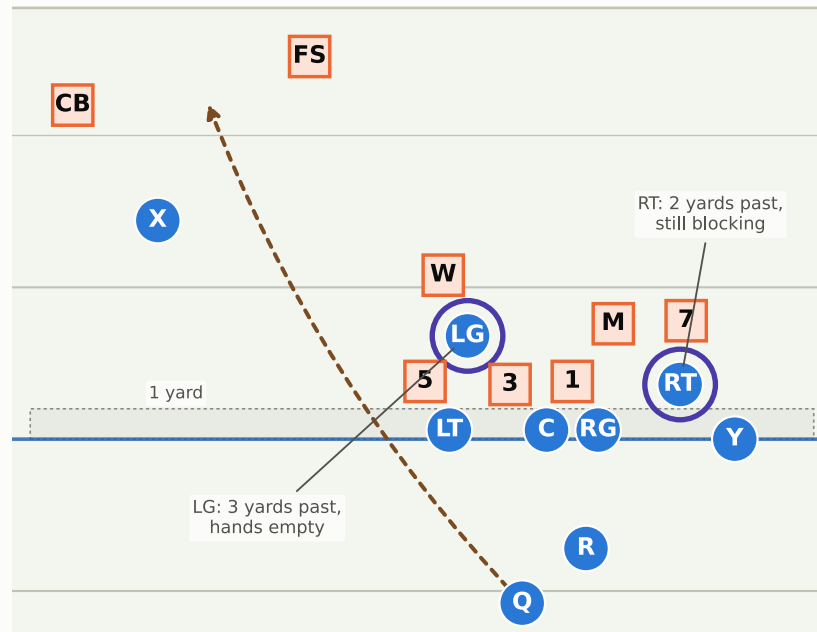


Figure 11: Drill 3: the instant of the throw. The left guard (ringed) is about three yards past the line of scrimmage with no defender in his hands; the right tackle (ringed) is two yards past it, still blocking the defensive end he engaged at the line. The pass is on its way to X.

Answer 3 is at the end of the chapter. Try it first.

Takeaways

- An **RPO** blocks a run and carries a pass on the same snap, and the quarterback picks one. Unlike **play-action**, the run is real; unlike the **option**, one branch is a pass. Every RPO is a **packaged play**.
- It attacks a **conflict defender** with a run job and a pass job, most often the **apex or overhang defender**, sometimes a linebacker or a safety. If he plays the run, throw behind him; if he plays the pass, hand off. The run is the default.
- **Pre-snap RPOs** (the **bubble RPO**) are decided by counting the box against the defenders outside; **post-snap RPOs** (the **glance**, **slant** and **stick RPOs** and the **pop pass**) are decided by watching one defender at the mesh. The read does a blocker's job.
- The NFL's **illegal man downfield** rule (one yard, with an exception for linemen still engaged with a defender they met at the line) makes NFL RPOs quick and short and keeps NFL linemen near the line; college's three yards past the neutral zone, for passes that cross it, lets college RPOs run-block longer and throw farther.
- RPOs grew out of Air Raid tags, the zone read and the college spread, and the Reid tree carried them to the NFL (the 2017 Chiefs and Eagles). In FTN's charting they are a few percent of NFL plays, mostly handoffs, with throws that cluster at or behind the line.
- Defenses answer by making the read lie (rotating a safety into the vacated run fit), playing two-high and fitting from depth, playing man to remove the conflict, stretching the clock to draw flags, and hitting the quarterback.

Connections

Builds on: Option Football (the read key, the mesh and give or keep), Play-Action, Bootlegs, and Screens (the fake that an RPO doesn't need), A First Look at Defense (the box and the apex), Zone Running (inside zone and the climb) and The Quick Game (the bubble, slant and stick).

Sets up:

- Run Fits: Gap Integrity, Force, Spill, and Box Math: how a defense fits the run against an RPO without losing the pass.
- Constraint Theory: RPOs as constraints, and how RPOs, play-action and the base run sequence off each other.
- Contact Rules and Penalties as Strategy: offensive pass interference and the other rules offenses and defenses play against.
- The College Laboratory: the Wing-T, the wishbone, the Air Raid and the spread option that produced the RPO.

See also (case studies): Andy Reid and the Kansas City Chiefs and The Eagles: Trenches, RPOs, and the Tush Push.

If you want to go deeper: Chris B. Brown's *The Art of Smart Football* and his Smart Football essays on packaged plays and the RPO; the RPO film breakdowns in USA Football's coaching blog and in college coaching clinics (the glance, bubble and slant RPOs are standard clinic topics); and, for the rules, the NFL's official playing rules (Rule 8, Sections 3 and 5) next to the NCAA rulebook (Rule 7-3), which make the one-yard and three-yard difference concrete.

Go deeper

From diagrams to tracking data. The animation in this chapter is a gridiron simulation written in NFL Big Data Bowl coordinates (simulated, BDB-format), so real tracking data drops into the same drawing code. The 2025 Big Data Bowl's `plays.csv` carries a PFF run-pass-option flag (`pff_runPassOption`), so finding real RPOs is a filter, not a hunt. With the tracking you can measure what this chapter only describes: how far past the line each lineman is at the frame of the `pass_forward` event, and how long the quarterback held the mesh. The cell below shows the shape of that code (it is not run, because the course does not download the Big Data Bowl files).

Answers to Predict the play

Answer 1.

Hand off, and the left side is the bonus.

- **The box:** four linemen and two linebackers, six defenders against six blockers (five linemen and the tight end). A hat on every hat: the run is fairly blocked.
- **Outside on the trips side:** H and Z against the nickel back, the cornerback and the strong safety, all within seven yards of the line. Two against three: the bubble is dead. The defense has done exactly what two thrown bubbles teach it to do.
- **The left side:** to get that third defender over the trips side *and* keep six in the box, the defense has left X alone against an off cornerback, with the only safety in the middle of the field. A slant or glance to X is the throw the count favors, if the play has one packaged to that side (many RPOs pair a bubble on one side with a slant or glance on the other, for exactly this picture).

The general lesson: the count is never just one side. A defense that wins the numbers in one place has lost them somewhere else, and a pre-snap RPO is a machine for finding where.

Answer 2.

Throw the slant. In six tenths of a second the safety has come two yards forward and two yards inside: shoulders toward the run, eyes in the backfield. He has chosen the run, and in doing so he has left the slant's landing spot, five yards past the line where he was standing. The quarterback pulls the ball and throws it on time, before the linemen have had a chance to drift past a yard. The cornerback, seven yards off and outside, can't get there. (Compare Figure 4: this is its upper branch.)

If the cornerback had lined up inside, the slant would have been a bad throw no matter what the safety did: the corner sits on the route's path. The quarterback would know that before the snap, and on a well-built call he'd have a second answer on that side (a quick out or a bubble) or he'd treat the play as a straight handoff. Leverage picks the live route; the read picks whether to throw it.

Answer 3.

NFL: yes, on the left guard; the right tackle is fine. Illegal man downfield, five yards from the previous spot, and the 15-yard completion comes back. The guard engaged the 3-technique within a yard of the line, but he *let go* and kept moving forward,

which the rule forbids once he's past a yard. The right tackle is legal under the rule's exception: he made contact at the line on his initial charge and has stayed locked on while driving his man downfield.

College: no foul. The limit is three yards beyond the neutral zone, and the guard is at about three yards past the line of scrimmage, inside the limit (barely; a step further and he wouldn't be). He hasn't reached the Will yet, which matters too: had he already been blocking him that far downfield, college could call offensive pass interference instead. The right tackle is legal under either rulebook.

This is the NFL RPO's whole problem in one frame. The guard did exactly what inside zone teaches, a double team and a climb to the backside linebacker. That's why NFL coaches either keep the guard on his double team when the play might be a pass, or read the Will instead of climbing to him, which is the glance RPO.